

RANBIR KUMAR

[+91 83760 86612](tel:+918376086612) · rk26.ftw@gmail.com · github.com/Ranbirkumar26 · linkedin.com/in/ranbir-kumar-a705551b5/

SUMMARY

CS (AI & ML) student at VIT Chennai with production experience in edge CV deployment and RAG-based semantic retrieval. Two papers under peer review in few-shot learning and terrain perception. SIH 2025 Finalist.

EDUCATION

B.Tech, Computer Science (AI & ML) — Vellore Institute of Technology, Chennai Aug 2023 – Present

CGPA: 8.22 / 10 | Expected Graduation: May 2027

Relevant Coursework: Machine Learning, Deep Learning, Natural Language Processing, Data Structures & Algorithms, Probability & Statistics, Computer Vision, Database Management (SQL)

EXPERIENCE

Computer Vision Engineer (Intern) | Proeffico Solutions Pvt. Ltd., Noida May – Jul 2025

- Designed and deployed 4 production CV pipelines (speed monitoring, pothole detection, fire detection, workplace safety analytics) on NVIDIA Jetson Xavier NX, achieving <40 ms real-time inference latency, reducing manual monitoring effort and enabling automated real-time decision-making in client workflows.

PROJECTS

Semantic Retrieval System; RAG Pipelines ([SemantiSearch](#)) *all-MiniLM-L6-v2 · GMM · FAISS · FastAPI · Docker*

- Built and containerized a semantic retrieval backend as a FastAPI/Docker microservice; validated end-to-end RAG pipeline with OpenAI achieving ~480 ms p95 latency, with confirmed LangChain/LlamaIndex compatibility.
- Implemented GMM-based cluster-partitioned semantic caching to group similar query embeddings, reducing redundant FAISS lookups by ~40% and improving average query response time.

Explainable Customer Intelligence System ([CraveConnect](#)) *Power BI · XGBoost · SHAP · LIME · Flask*

- Built a Power BI-integrated end-to-end ML inference system with SHAP and LIME explainability, enabling interpretable predictions and decision-driven customer segmentation via a deployed Flask API.

[Annadata](#) – Agricultural AI Platform (SIH 2025 Finalist) *YOLOv8 · ResNet-34 · Flask · Python*

- Developed an AI agritech platform using YOLOv8 and ResNet-34, achieving 91% disease classification and 97% soil health prediction accuracy, with GPS-enabled asynchronous pipelines for real-time field monitoring.

RESEARCH and EXTRACURRICULUM

Few-Shot Learning with Explainable AI *Peer-reviewed journal, under review*

- Designed N-way K-shot classifier on hospital-sourced low-data medical diagnosis; achieved 93% diagnostic accuracy with LIME + SHAP attribution for full prediction interpretability.

Vision-Based Terrain Analysis *Peer-reviewed journal, under review*

- Developed real-time multi-model perception system (YOLOv8 + MiDaS) for terrain classification in autonomous navigation; deployed to embedded edge hardware.

Team Lead (Rover) | Technocrats Robotics, VIT Chennai *Mar 2024 – Present*

- Architected autonomous terrain analysis system integrating CV-based soil classification and sensor fusion, contributing to All-India Rank 1 (Robocon Round 1) and IRC 2024 finals.

SKILLS

AI / ML: RAG Pipeline Design, Semantic Search, Vector Embeddings, Explainable AI (SHAP, LIME), Few-Shot Learning, Prompt Engineering, LLM Integration (LangChain, LlamaIndex)

Computer Vision: YOLOv8, ResNet, MiDaS, Object Detection, Depth Estimation, Real-Time Inference, Edge Deployment (NVIDIA Jetson Xavier NX)

ML Frameworks: PyTorch, Scikit-learn, XGBoost, HuggingFace Transformers, FAISS

Backend & Deployment: FastAPI, Flask, Docker, REST APIs, Microservices

Data & Tools: Power BI (Dashboarding, DAX, Data Modeling, Marketing Analytics), Pandas, NumPy, SQL, Git, Linux

ACHIEVEMENTS

- SIH Finalist (2025 Hardware) | All-India Rank 1 DD Robocon | Finalist International Rover Challenge (IRC-2026)